

Section 12.1 (Three Dimensional coordinate system)

Basic definitions in 3-dimensional rectangular coordinate system

- **The Cartesian product**

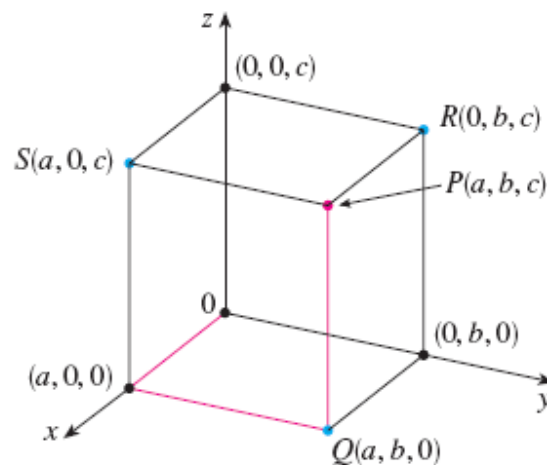
The Cartesian product $\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \{(x, y, z) : x, y, z \in \mathbb{R}\}$

is the set of all ordered triples of real numbers and is denoted by $\mathbb{R}^3$.

The one-to-one correspondence between points P in the space has an ordered triple (a, b, c) in $\mathbb{R}^3$:

Any point P in the space has an ordered triple (a, b, c) in $\mathbb{R}^3$, and it means:

To locate P , we start from the origin, move a -units along X -axis, then b -units parallel to Y -axis and then c -units parallel to Z -axis.



Remark:

The point $P(a, b, c)$ determines a rectangular box as in Figure 5. If we drop a perpendicular from P to the xy -plane, we get a point Q with coordinates $(a, b, 0)$ called the **projection** of P on the xy -plane. Similarly, $R(0, b, c)$ and $S(a, 0, c)$ are the projections of P on the yz -plane and xz -plane, respectively.

Basic definitions in 3-dimensional rectangular coordinate system (Cont...)

- Coordinate Axes**

Three mutually perpendicular coordinate lines **X-axis, Y-axis, Z-axis** (intersecting at origin).

- Coordinate Planes**

Three planes determined by coordinate axes **XY-plane, XZ-plane, and YZ-plane.**

- Octants**

Coordinate planes divide 3-space into 8 parts called octants.

- Planes**

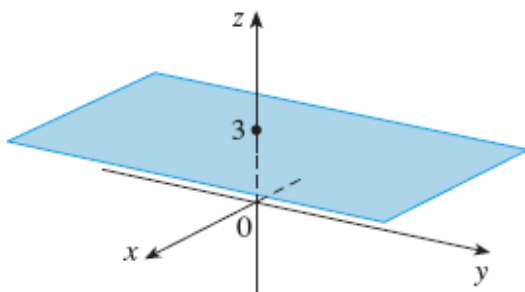
EXAMPLE I What surfaces in $\mathbb{R}^3$ are represented by the following equations?

(a) $z = 3$

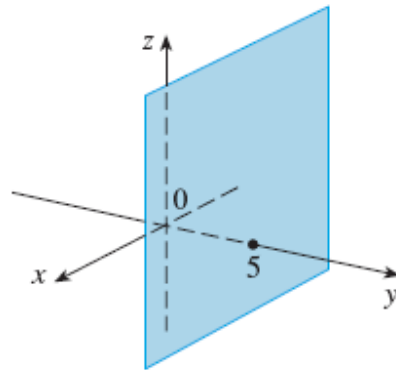
(b) $y = 5$

Solution

Done in class.



(a) $z = 3$, a plane in $\mathbb{R}^3$



(b) $y = 5$, a plane in $\mathbb{R}^3$

Note:

In general, if k is a constant, then $x = k$ represents a plane parallel to the yz -plane, $y = k$ is a plane parallel to the xz -plane, and $z = k$ is a plane parallel to the xy -plane.

Exercise

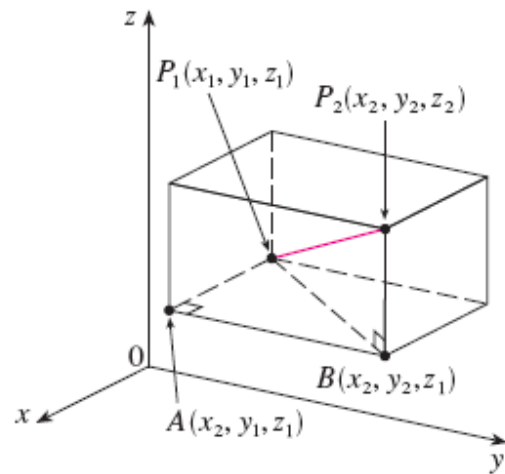
Describe and sketch the surface in $\mathbb{R}^3$ represented by the equation $y = x$.

Basic formulas in 3-dimensional coordinate system

Distance Formula

The distance between points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



Example

Find the distance between $P_1(2, -1, 7)$ and $P_2(1, -3, 5)$.

Answer: 3

Midpoint Formula

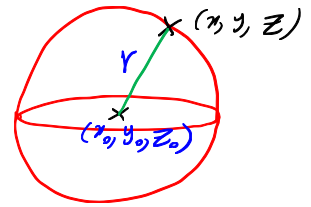
The coordinates of midpoint of $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ are

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

Example

Find the midpoint of $P_1(2, -1, 7)$ and $P_2(1, -3, 5)$.

Answer: $\left(\frac{3}{2}, -2, 6\right)$



Simple graphs in 3-space

• Spheres

Sphere with center (x_0, y_0, z_0) and radius r is given by

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = r^2$$

Need

- center
- radius

By completing
the square

Given any equation of the form

$$x^2 + y^2 + z^2 + Gx + Hy + Iz + J = 0.$$

We can write it as

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = k$$

If $k > 0$
then sphere

If $k = 0$
then only the point
 (x_0, y_0, z_0)

If $k < 0$
then no graph

Example

What is the graph of $2x^2 + 2y^2 + 2z^2 + 8x + 12y + 18 = 0$?

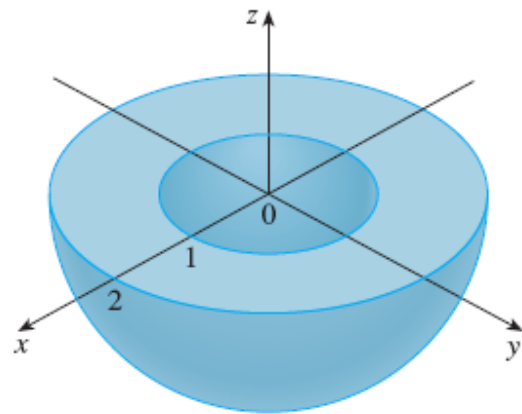
Solution

Done in class.

Simple graphs in 3-space

Example What region is represented by $1 \leq x^2 + y^2 + z^2 \leq 4, z \leq 0$?

Solution Done in class.

**Exercise**

Find the center and radius of the sphere that has $(1, -2, 4)$ and $(3, 4, -12)$ as end points of a diameter. Give the equation of the sphere.

End of Section 12.1